Jenkins Slave Configuration

TCP Connection- over TCP port

Two methods to achieve Jenkins slave configuration:

1. **SSH** - Port 22
2. **JNLP** - Java Network Launch Protocol - with any random TCP port (0 - 65535)

SLAVE / NODE / AGENT / LABEL - all are same

Dashboard → Manage Jenkins → Nodes
You can see all your nodes here,
Jenkins master - Build-In Node

Let's launch and prepare a slave machine: (slave-1)

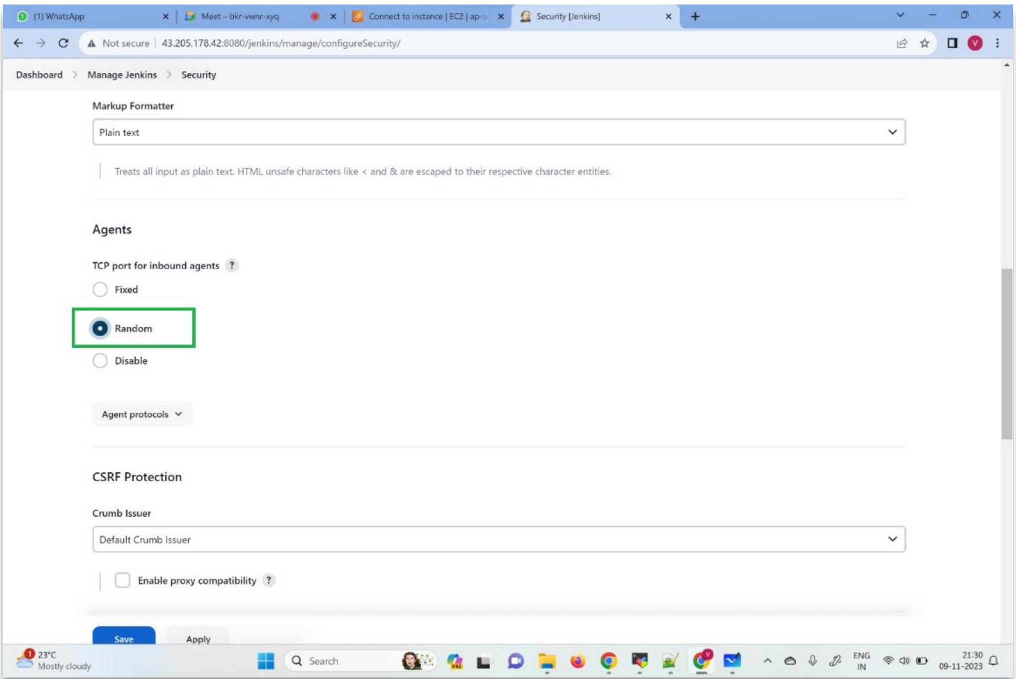
Launch an Ec2 instance → name (slave-1) → login to machine →
```
$ sudo su -
# amazon-linux-extras install java-onpejdk11 -y
```
(install java on slave machine *must have java on slave machine)

Create one directory as a slave's remote root directory where we will download our agent.jar file later in this practical.

Eg:
```
cd /mnt
mkdir jenkins-slave
```

JNLP Method:

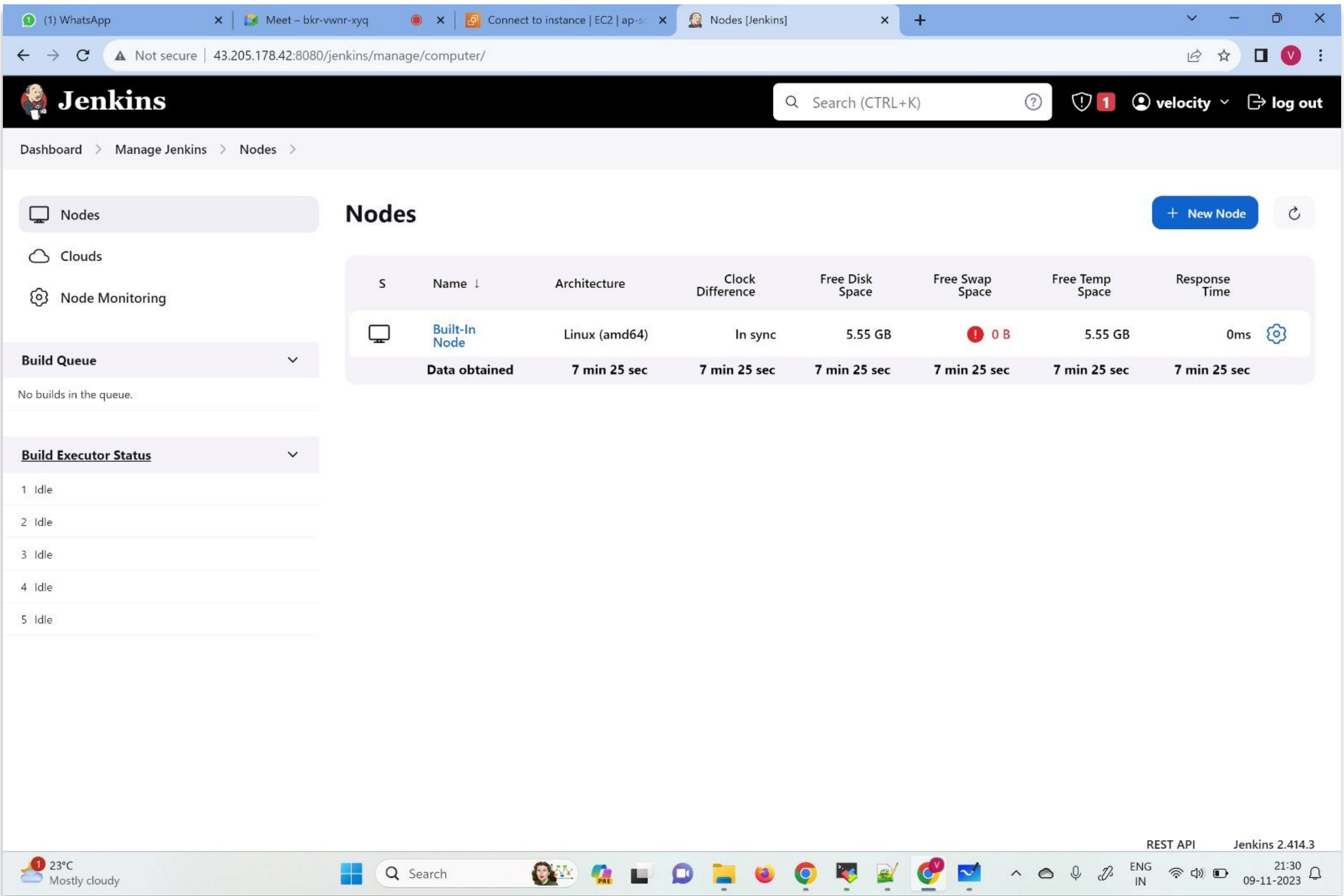
Go to Manage Jenkins → Security → Agents → select TCP port for inbound agents as **Random** (we will make it **Fixed** later)



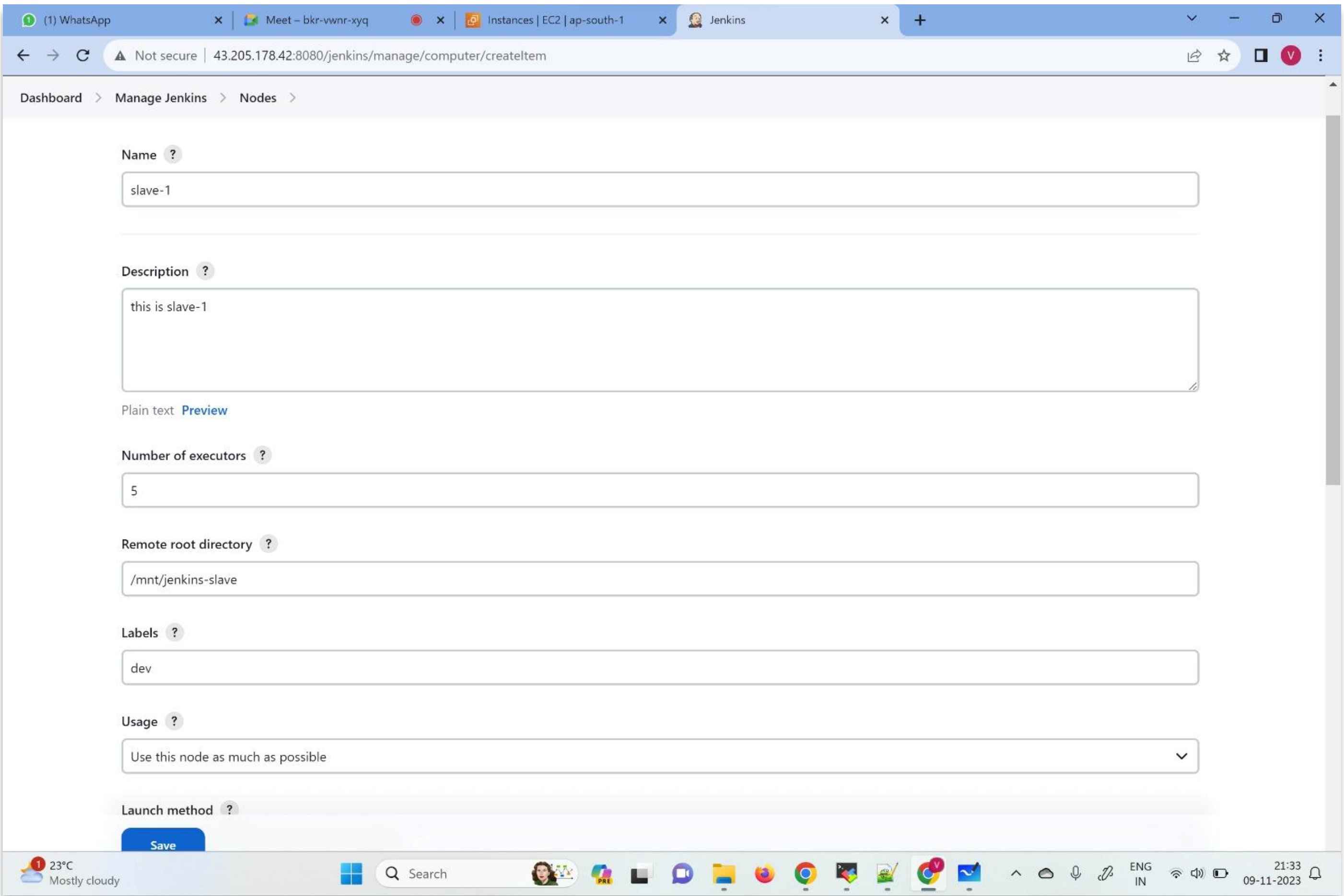
→ **Apply + Save**

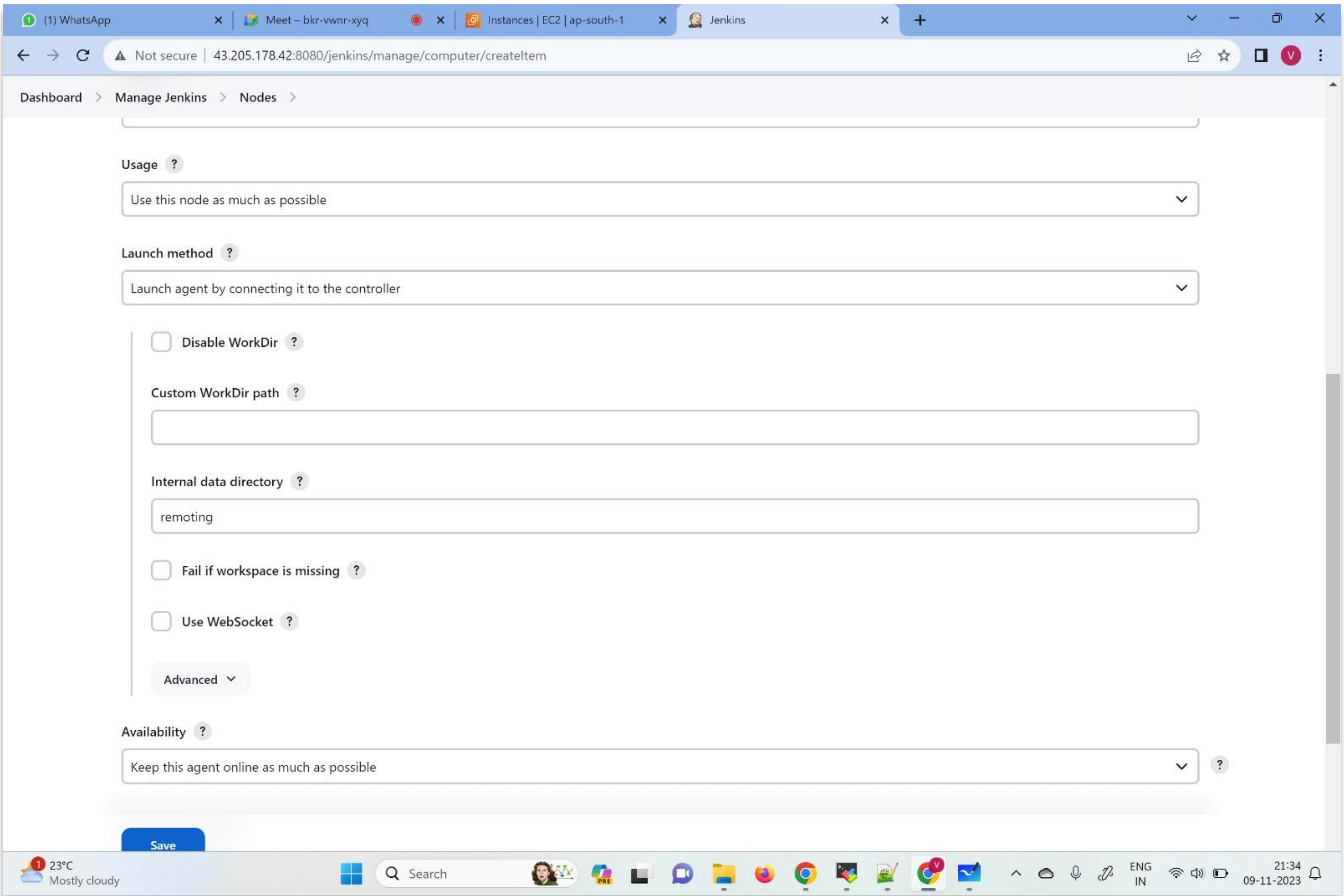
We will create a slave now

Dashboard → Manage Jenkins → Nodes →



Click on New node → add node name (slave-1) → select Type Permanent Agent → click Create → add description → Number of executors (eg: 5) → add Remote root directory (Eg: /mnt/jenkins-slave ; this directory has to be present on machine Jenkins will not create for us) → add label (must have, eg: dev) → Launch method → Launch agent by connecting it to the connector (this is JNLP) → click on **save**


(1) WhatsApp
Meet – bkr-vwnr-xyq
Instances | EC2 | ap-south-1
Jenkins
Not secure | 43.205.178.42:8080/jenkins/manage/computer/createItem
Dashboard > Manage Jenkins > Nodes >
Name ?
slave-1
Description ?
this is slave-1
Plain text Preview
Number of executors ?
5
Remote root directory ?
/mnt/jenkins-slave
Labels ?
dev
Usage ?
Use this node as much as possible
Launch method ?
Save
23°C
Mostly cloudy
Search
ENG
IN
21:33
09-11-2023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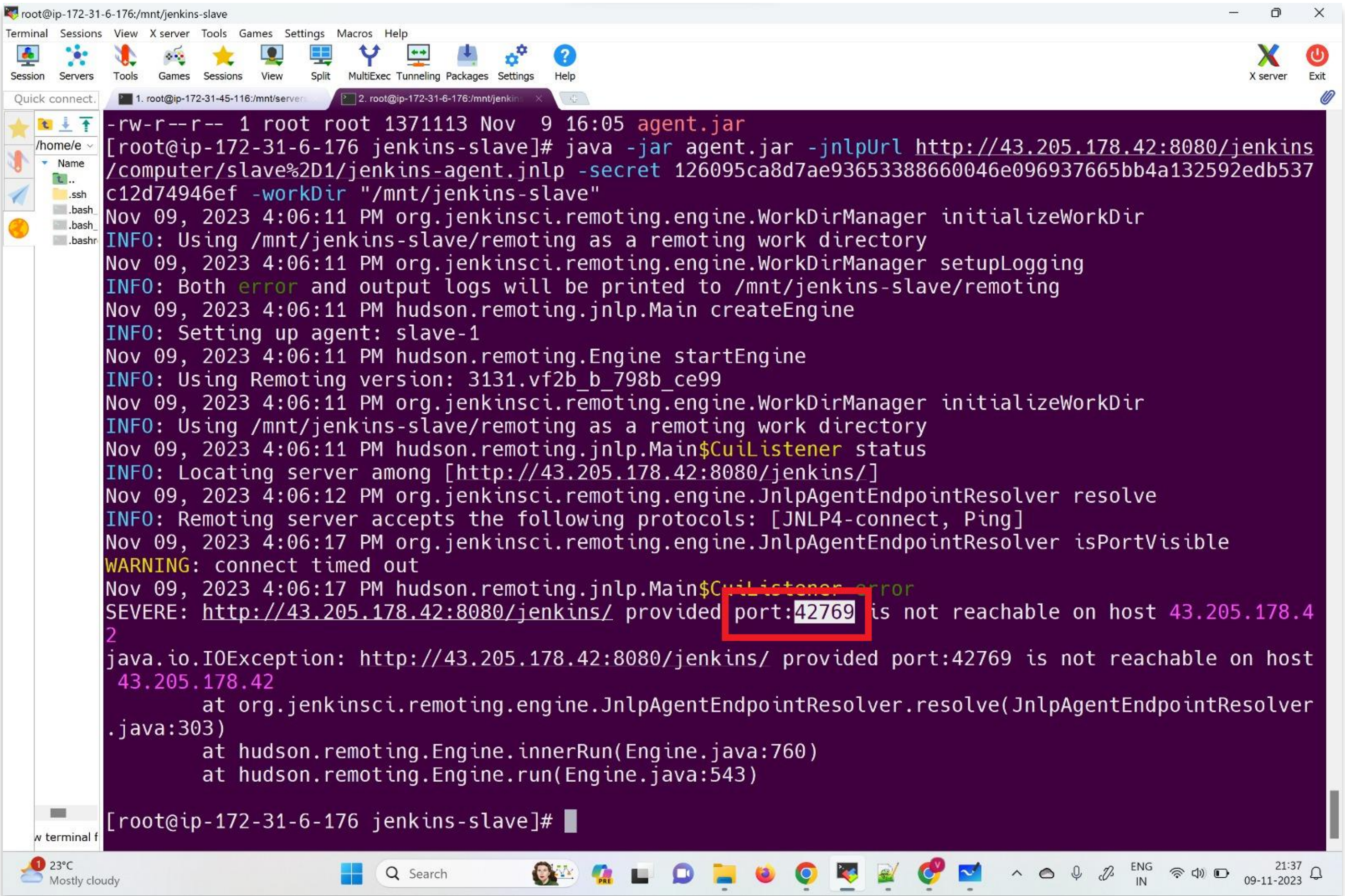
Jenkins
Not secure | 43.205.178.42:8080/jenkins/manage/computer/createItem
Dashboard > Manage Jenkins > Nodes >
Usage ?
Use this node as much as possible
Launch method ?
Launch agent by connecting it to the controller
Disable WorkDir ?
Custom WorkDir path ?
Internal data directory ?
remoting
Fail if workspace is missing ?
Use WebSocket ?
Advanced
Availability ?
Keep this agent online as much as possible
Save
23°C
Mostly cloudy
Search
ENG
IN
21:34
09-11-2023

You can see new slave-1 is created but it's disconnected, now open slave-1 node you will see a few commands. We will connect our slave to master with the help of this commands: (check if Jenkins URL is updated by going to - Dashboard – Manage Jenkins – System – Jenkins URL)

Now download agent.jar on slave machine, inside slave's Remote root directory (Eg: /mnt/jenkins-slave) using curl command.

Again, run the second command starting with - java -jar ……. … …

It will try to connect but will give an error, in the logs we can see why it failed to connect over a specific port number. The reason is that the port number is not open in our machine's security group from where it will attempt to connect with our Master. we can also add this port number in Jenkins security and make it fixed.



For that go to, Dashboard → Manage Jenkins → Security → Inside Agents → tcp port for inbound → select Fixed and add port number here.

We need to open the same port on the machine's security group then run the same command again and we can see that our slave will now connect to our Master.

Check by going to → Dashboard → Manage Jenkins → Nodes → slave–1

If we press CTRL+C on terminal it will lose the connection with master, hence to run the command in background Use same command starting with 'nohup' and ending with '&'

Eg: `# nohup <command> &`

Run the command now and press enter key 2 - 3 times. our command is running in the background.

Command History of Slave Machine

```
# amazon-linux-extras install java-openjdk11
# cd /mnt
# mkdir jenkins-slave
# cd jenkins-slave/
# curl -sO http://43.205.178.42:8080/jenkins/jnlpJars/agent.jar
# nohup java -jar agent.jar -jnlpUrl
http://43.205.178.42:8080/jenkins/computer/slave%2D2/jenkins-agent.jnlp
-secret
07fea906afed6aaca3d88a5338c67b6d50b9bc31eb40e4d13908592aaa658cf8 -
workDir "/mnt/jenkins-slave" &
tail -100f c
tail -100f nohup.out
```

Now, create one more machine (slave-2) by repeating above steps.

Lets run a job:

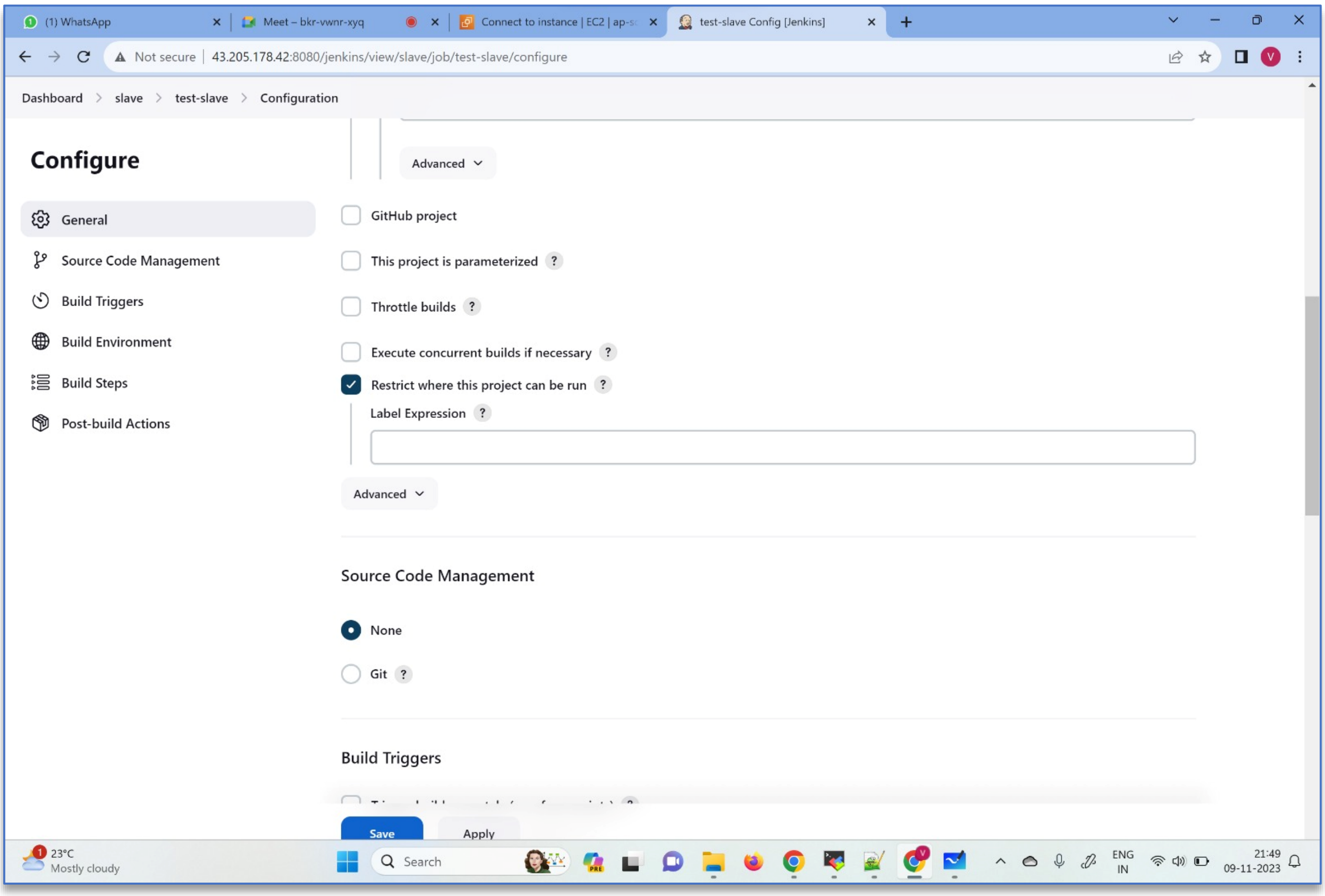
New Item → add name (test-slave) → Freestyle Project → OK → Add build steps → execute shell

```
echo “hello!!!”
```

→ select Delete workspace before build start → Discard Old builds → **Apply + Save**

We haven't mentioned anything about slaves related to this job so by default Jenkins will run jobs on any available free slaves.

To specify where our job should run:



Open Job and go to configure → select restrict where this project can be run → mention Label Expression > add ‘dev’ (it will run job on any free machine which has label dev) → **Apply + Save**

Let's say if we want to run job on particular slave - just write name of the slave instead of label (Eg: slave-1) next time when you build a job it will execute on that particular machine.